

Engineering Reference Document

[11] [19] [45] Date Compiled

ECX13010-R2
Sept. 12, 2026

ECX13010 Interface Study

- [54] **AUTOMOTIVE-STYLE ESC / RECEIVER INTERFACE STUDY**
- [75] **Subject Unit:**
ECX13010, integrated 2-in-1 ESC / receiver
- [73] **Platform:**
ECX V4 chassis, 2.4 GHz radio system
- [21] **Ref. No.:** ECX13010-R2
- [22] **Compiled:** Sept. 12, 2026
- [30] **Prior Documentation on File**
Sept. 12, 2026 [internal] None — first structured pass
- [51] **Category** RC electronics — ESC/receiver combo
- Function**
- [52] **Class** 20 A ESC; integrated 2.4 GHz Rx
- Field of**
- [58] **Reference** RC drivetrain electronics; embedded control interfacing

Reviewed by: — (unreviewed draft)
Prepared for: ESP32-C3 integration research

[57] ABSTRACT

This document describes the ECX13010 integrated electronic speed controller and 2.4 GHz receiver. It records manufacturer-documented characteristics, observed connector labels, the suspected purpose of each observed connection, and a controlled approach for investigating compatibility with an ESP32-C3. Confirmed specifications are deliberately separated from field observations that still require measurement before any control signal is applied.

9 Sections, 1 Figure, 3 Reference Tables

[56] References Cited

Manufacturer-Documented Specifications

Parameter	Documented Value
ESC rating	20 A
BEC output	6 V / 2 A
Radio	2.4 GHz, ECX V4
Architecture	2-in-1 ESC / receiver

User-Observed Labels (Unverified)

Label	Location	Status
3CH	Channel header	Observed; function unconfirmed
3CH1 / 3CH2 / 3CH3	Channel header	Observed; function unconfirmed
3	Suspected VCC	Requires voltage measurement
2	Suspected GND	Requires continuity check

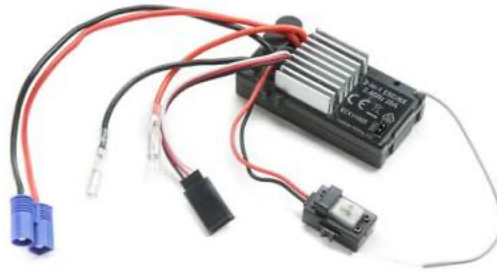


FIG. 1 — ECX13010 reference photograph, as received.

1. DEVICE IDENTIFICATION

Part number: ECX13010 **Function:** ESC + receiver **Radio:** ECX V4, 2.4 GHz **Application:** ECX 1/10-scale RC systems

2. DOCUMENTED CHARACTERISTICS

Parameter	Documented Value
ESC rating	20 A
BEC	6 V / 2 A
Radio	2.4 GHz V4
Architecture	2-in-1 ESC / receiver

3. OBSERVED PIN LABELS

The following labels were reported from inspection of the user's unit: **3CH**, **3CH1**, **3CH2**, **3CH3**, **3**, and **2**. The roles below are presented as engineering hypotheses, since no official pin-level schematic was available at the time of writing.

Label	Proposed Role	Interpretation	Confidence
3CH	Channel 3 / auxiliary	Potential auxiliary receiver-control connection; signal must be identified.	Unverified
3CH1	Channel-related	Potential channel interface. Do not assume GPIO/PWM/UART without measurement.	Unverified
3CH2	Channel-related	Potential channel interface; direction and voltage require measurement.	Unverified
3CH3	Channel-related	Potential channel interface; exact protocol requires measurement.	Unverified
3	Suspected VCC	May be a supply rail if voltage measurement confirms it.	Unverified
2	Suspected GND	May be ground if continuity and voltage tests confirm it.	Unverified

4. HOW THE CONNECTIONS FIT INTO THE SYSTEM

Power and ground: a control circuit requires a supply rail and a reference ground. If pins 3 and 2 are confirmed as VCC and GND on this particular unit, compatible electronics could use those rails within their electrical limits — but their identities must be measured first.

Channel interfaces: a receiver channel carries control information associated with a transmitter function. Channel numbering alone does not establish a specific electrical protocol. The ECX13010 is an integrated receiver/ESC, so its internal header arrangement should not be assumed to match a generic stand-alone receiver.

3CH / 3CH1 / 3CH2 / 3CH3: these markings indicate channel-related connections based on their labels, but the exact pin-by-pin electrical behavior is not established here.

5. ESP32-C3 INTEGRATION MODEL

A safe research architecture is:

ESP32-C3 → verified level-compatible interface → ECX13010 control input → ESC power stage → motor

The ESP32-C3 uses 3.3 V GPIO. An unidentified ECX13010 pin should not be connected directly to an ESP32 GPIO. Establish voltage, signal type, signal direction, and electrical limits first.

6. PIN CHARACTERIZATION PROCEDURE

For each candidate connection, record the following before attempting control:

Test	Record
Physical label / location	_____
Voltage to confirmed ground	_____ V
Logic-high voltage	_____ V
Logic-low voltage	_____ V
Signal frequency	_____ Hz
Signal type	PWM / serial / other / unknown
Signal direction	Input / output / unknown
ESP32 compatibility	Confirmed / not confirmed

7. FUNCTIONAL BLOCK DIAGRAM

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2.4 GHz transmitter
↓
ECX13010 receiver / control logic
↓
ESC power stage
↓
Motor

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8. ENGINEERING SAFETY

Disconnect the battery before changing wiring. Do not treat a visual pin interpretation as a confirmed schematic. Verify voltage and signal behavior with appropriate test equipment before attaching an ESP32-C3. Initial motor tests should be performed with the mechanical system secured and with no condition that could cause an unexpected motor movement to injure someone.

9. CONCLUSION

The ECX13010 is a 2.4 GHz V4 integrated ESC/receiver. Its documented system characteristics can be recorded with confidence, while the observed channel and power labels in this document remain provisional until measured. This distinction matters when converting the RC component into an experimental robotics system without relying on an unverified pinout.

DOCUMENT CONTROL

Device: ECX13010 Revision: 2.0 Purpose: robotics interface research Classification: preliminary engineering reference, formatted for internal use only.